Nowcasting

Instructions

Workshop 2 – Workshop on Useful Statistical Procedure for Nowcasting Part 2: ARMA Models



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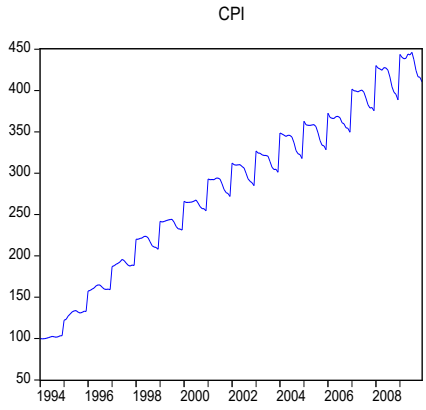
At the end of this workshop you should be able to:

- Estimate an ARMA(p,q) model;
- Work through the Box-Jenkins methodology for determining the appropriate values for p and q;
- Generate forecasts for an ARMA model.

The dataset for this workshop can be found in the page "Chile" of the work file "**arma_data.wf1**". We will be using the series CPI, which measures the price level of Chile for the period 1994M1 to 2009M12.

Question 1. Graph the data in the CPI series. Based on visual inspection, discuss the stationarity of the *cpi* series and any other salient feature that you notice.

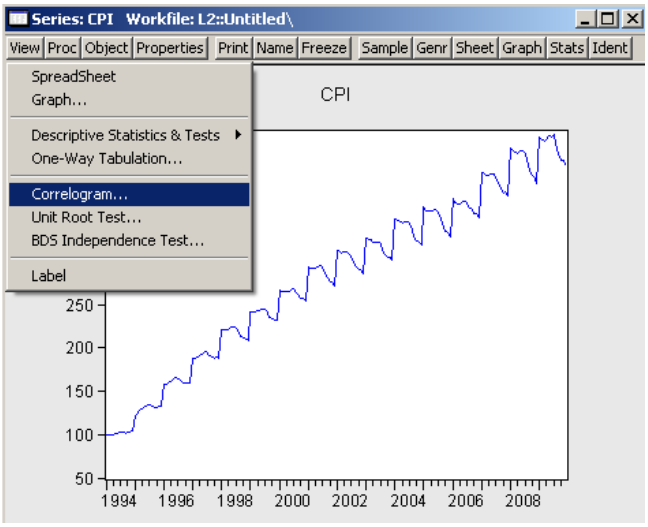
To do so, double click on the CPI series object in the work file window and then choose **View-Graph-Ok.**

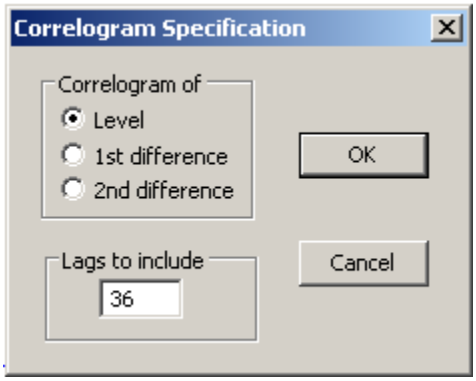


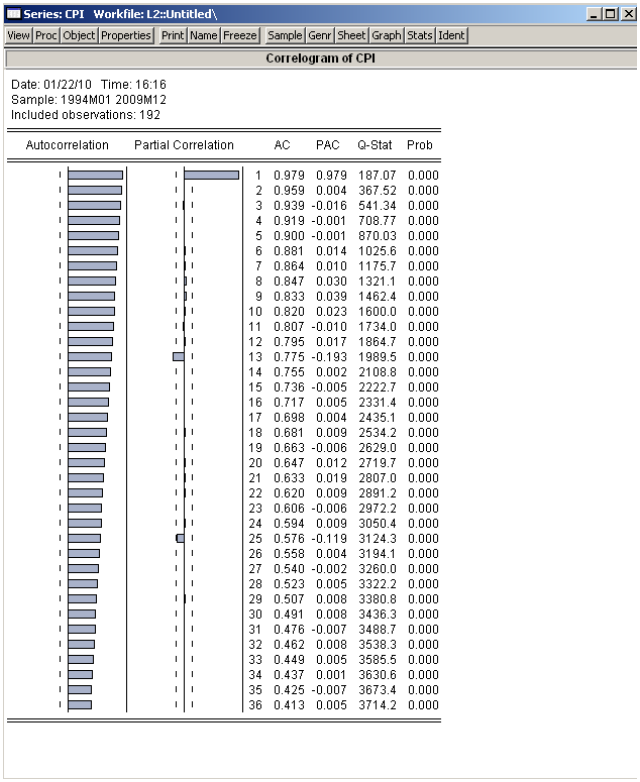
The CPI series is clearly not stationary: both the mean and variance increase over time. Also, there is obvious seasonality in the CPI series.

Question 2. Calculate the ACF and PACF of the CPI series. What can you infer from the ACF/PACF about the structure of the ARMA model to predict CPI?

Choose **View-Correlogram…**
Choose **36** lags and **Level**. In this case we'll see the ACF and the PACF up to 36 lags.
Choose OK.





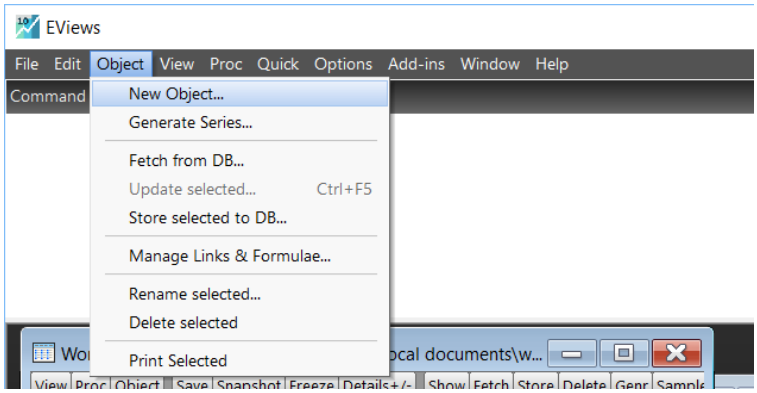


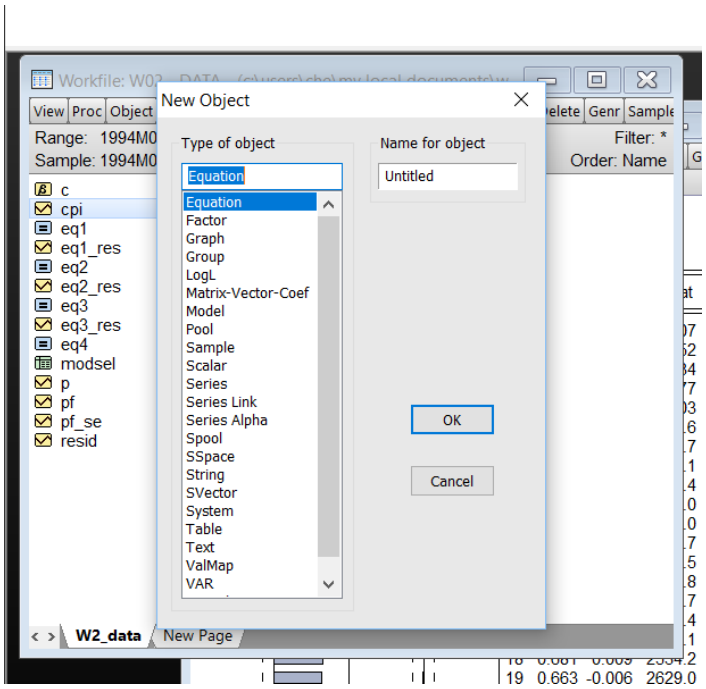
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.979	0.979	187.07	0.000
		2	0.959	0.004	367.52	0.000
		3	0.939	-0.016	541.34	0.000
		4	0.919	-0.001	708.77	0.000
		5	0.900	-0.001	870.03	0.000
		6	0.881	0.014	1025.6	0.000
		7	0.864	0.010	1175.7	0.000
		8	0.847	0.030	1321.1	0.000
		9	0.833	0.039	1462.4	0.000
		10	0.820	0.023	1600.0	0.000
		11	0.807	-0.010	1734.0	0.000
		12	0.795	0.017	1864.7	0.000
		13	0.775	-0.193	1989.5	0.000
		14	0.755	0.002	2108.8	0.000
		15	0.736	-0.005	2222.7	0.000
		16	0.717	0.005	2331.4	0.000
		17	0.698	0.004	2435.1	0.000
		18	0.681	0.009	2534.2	0.000
		19	0.663	-0.006	2629.0	0.000
		20	0.647	0.012	2719.7	0.000
		21	0.633	0.019	2807.0	0.000
		22	0.620	0.009	2891.2	0.000
		23	0.606	-0.006	2972.2	0.000
		24	0.594	0.009	3050.4	0.000
		25	0.576	-0.119	3124.3	0.000
		26	0.558	0.004	3194.1	0.000
		27	0.540	-0.002	3260.0	0.000
		28	0.523	0.005	3322.2	0.000
		29	0.507	0.008	3380.8	0.000
		30	0.491	0.008	3436.3	0.000
		31	0.476	-0.007	3488.7	0.000
		32	0.462	0.008	3538.3	0.000
		33	0.449	0.005	3585.5	0.000
		34	0.437	0.001	3630.6	0.000
		35	0.425	-0.007	3673.4	0.000
		36	0.413	0.005	3714.2	0.000

Given that the ACF decrease very slowly, perhaps the MA component of the CPI series has a large "q". However, the parsimony principle suggests that it is better to estimate an AR(1) instead of a MA(∞) model.

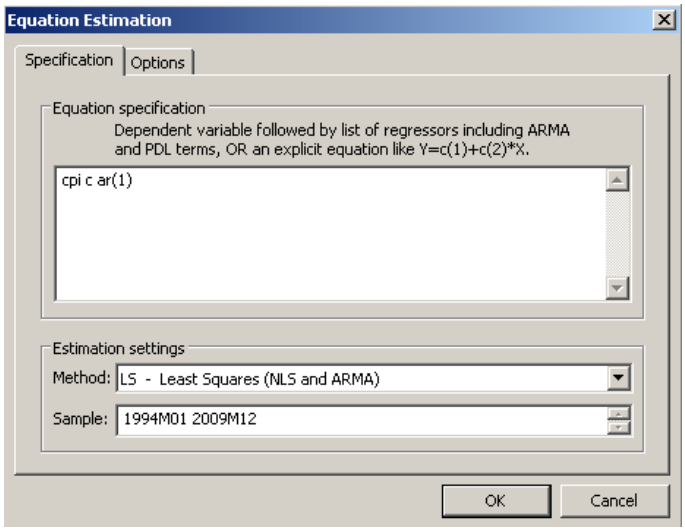
Question 3. Estimate an AR(1) model for the CPI series. Examine the stationarity of the estimated model by calculating the roots of the AR polynomial. Assess the presence of seasonality by examining the residuals visually. Calculate the ACF and PACF of the estimated model residuals.

To do so, first in the Eviews **Object** menu choose **New Object**, in the type of object choose Equation, choose Name for object and click ok. (equivalently, **Quick** menu choose **Estimate Equation**.)

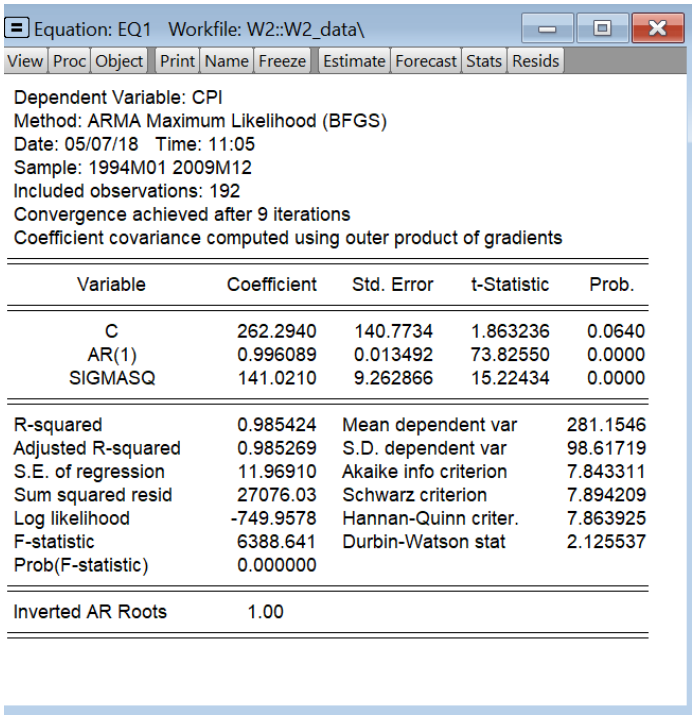




Then write “cpi c ar(1)” in the “Equation Specification” box (“c” means including a constant).

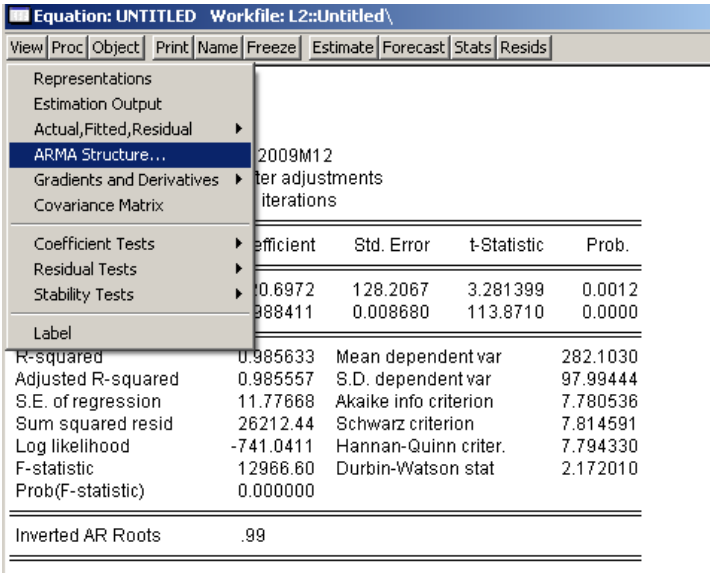


And then press OK. (In different versions of Eviews the estimation output may look slightly different due to changes in default settings, but it won't change our conclusions at all.)

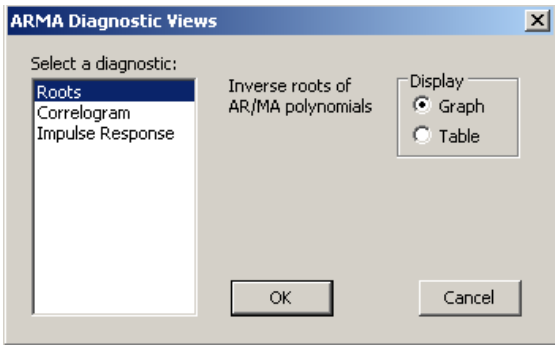


Next, we analyze the Stationarity of the model.

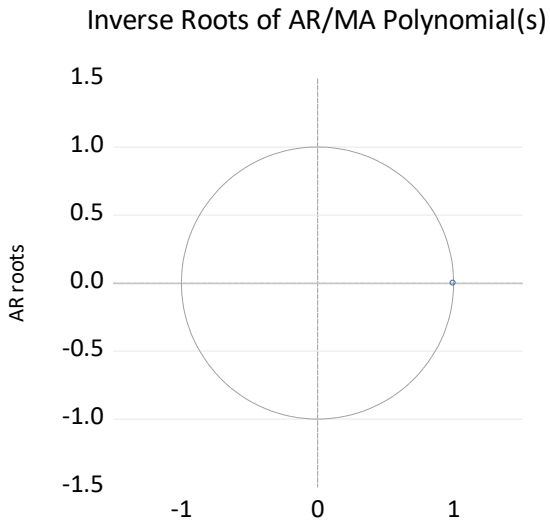
First Choose **View --> ARMA Structure**:



Then follow the defaults. First choose ROOTS – GRAPH and OK.



We can see that the root is smaller but very close to 1. It seems stationary but should worry us (More on this in Lecture 3).



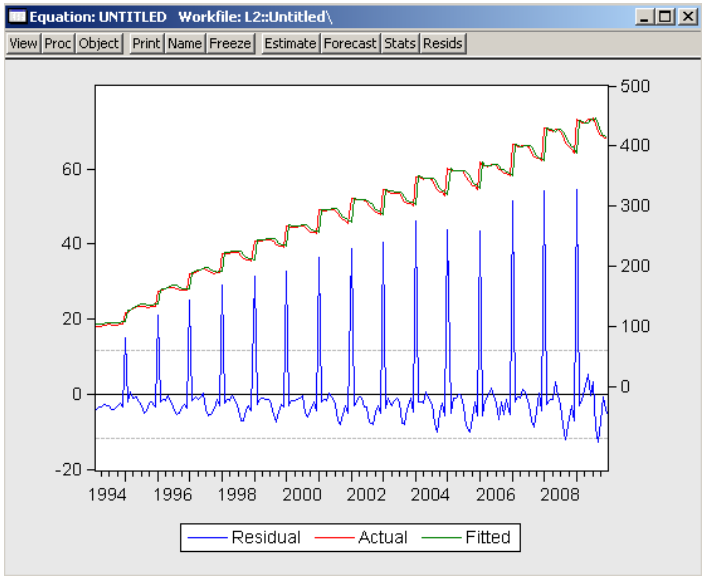
If you instead choose TABLE you get the root and the modulus, which is the same here because the root is rational, not complex (i.e., imaginary). The root equals the coefficient of AR(1), why?

Inverse Roots of AR/MA Polynomial(s)
Specification: CPI C AR(1)
Date: 01/12/20 Time: 20:29
Sample: 1994M01 2010M12
Included observations: 192

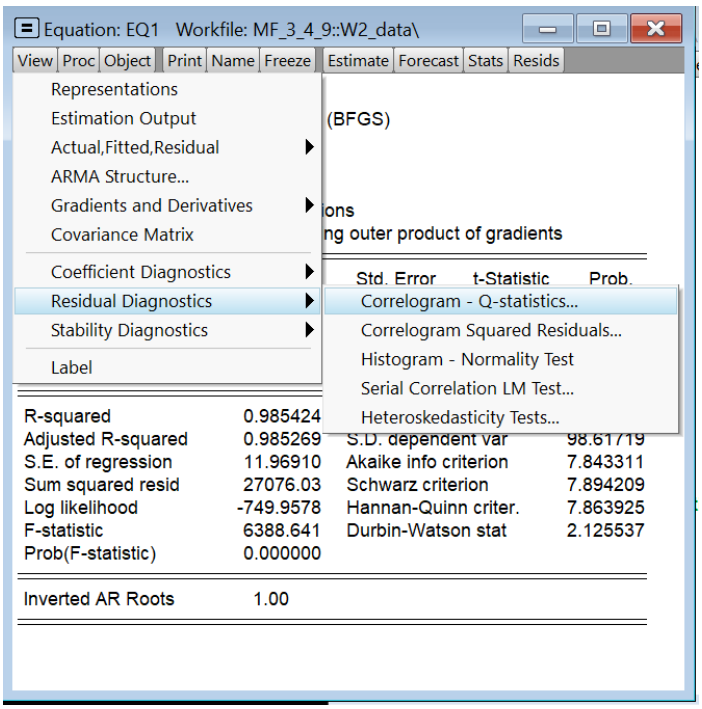
AR Root(s)	Modulus	Cycle
0.996089	0.996089	

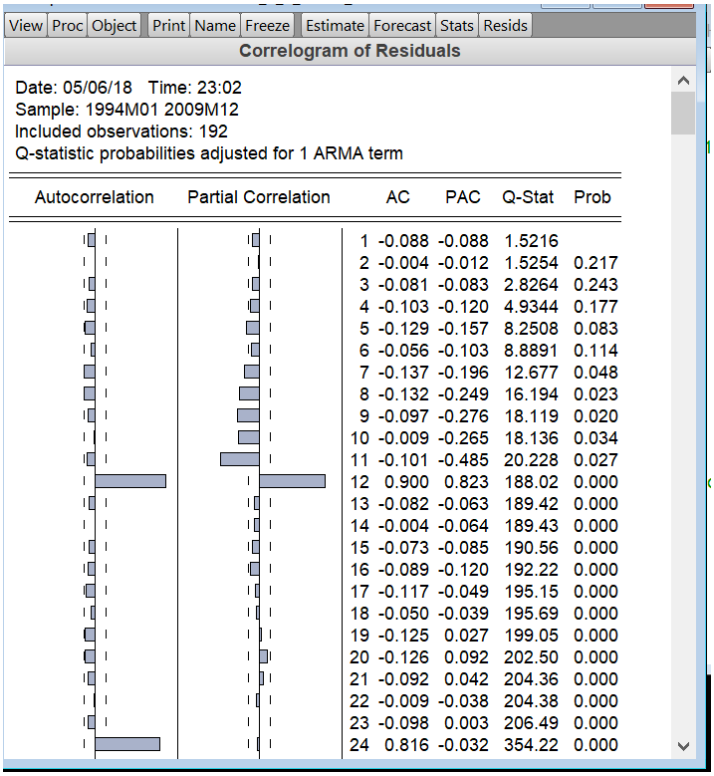
No root lies outside the unit circle.
ARMA model is stationary.

Looking at the Residuals (Choose Tab "**Resids**") we can see the strong seasonality in the CPI.



Which is confirmed by the correlogram of the residuals (note the peaks of the ACF at multiples of 12). Choose **View → Residual Diagnostics → Correlogram – Q-statistics** and click **Ok** to choose the default lags.

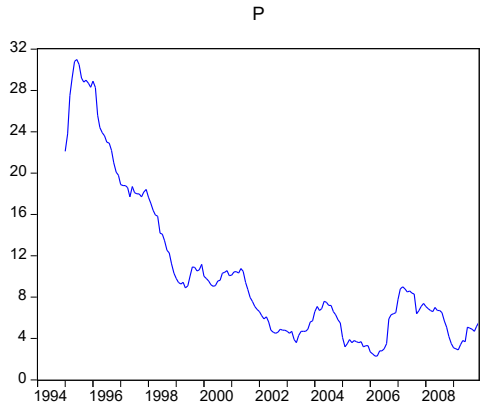




The spike at lag 12 and 24 indicates seasonality.

Question 4. Calculate the 12 month inflation rate as p = ((cpi/cpi(-12))-1)*100. Inspect visually its graph. Is something happening around 2000? Is there a structural break? (Extra info: Chile introduced inflation targeting at the end of 1999.)

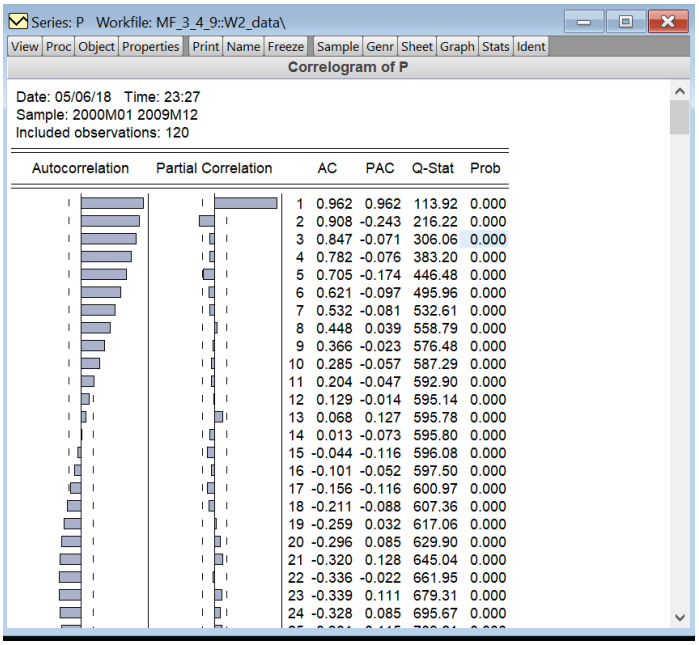
To do so, go back to the cpi object window, choose **Proc → Generate by equation**, type "p = ((cpi/cpi(-12))-1)*100" in the **enter equation box** and **click ok**. Back to the work file window a new series object p is generated. Double click and choose **View->Graph, OK**



After a disinflationary period during the 1990s, Chile introduced inflation targeting from 2000. Let's constrain our sample to after 2000.

Question 5: Repeat Question 2 and 3 for inflation (i.e., the series labeled P) <u>focusing on the post 2000 sample only</u>. Experiment with an AR(1) and then an AR(2) model for P, and decide between these two models using the AIC-HQ-BIC statistics.

To choose sample, click Sample and type "2000M1 @last" or "2000M1 2010M12" in the Sample box.



For P (yoy inflation) there is no evidence of seasonality, and the PACF indicates an AR(1) or AR(2), while ACF indicates MA(high), meaning AR(1) at least.

Let's estimate an AR(1). Choose **Object→ New Object → Equation, OK** and in the equation specification type "p c ar(1)":

Equation: EQ2 Workfile: MF_3_4_9::W2_data\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: P
Method: ARMA Maximum Likelihood (BFGS)
Date: 05/06/18 Time: 23:37
Sample: 2000M01 2009M12
Included observations: 120
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.815559	1.363862	4.997249	0.0000
AR(1)	0.976046	0.022066	44.23386	0.0000
SIGMASQ	0.276733	0.024516	11.28800	0.0000

R-squared	0.948566	Mean dependent var	6.140182
Adjusted R-squared	0.947687	S.D. dependent var	2.329278
S.E. of regression	0.532755	Akaike info criterion	1.628594
Sum squared resid	33.20791	Schwarz criterion	1.698282
Log likelihood	-94.71566	Hannan-Quinn criter.	1.656895
F-statistic	1078.876	Durbin-Watson stat	1.310594
Prob(F-statistic)	0.000000		

Inverted AR Roots	.98

And look at the correlogram of the residuals:

Equation: EQ2 Workfile: MF_3_4_9::W2_data\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Correlogram of Residuals

Date: 05/06/18 Time: 23:39
Sample: 2000M01 2009M12
Included observations: 120
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.328	0.328	13.274	
		2	0.135	0.031	15.547	0.000
		3	0.129	0.085	17.615	0.000
		4	0.229	0.181	24.250	0.000
		5	0.137	0.005	26.636	0.000
		6	0.018	-0.062	26.678	0.000
		7	-0.100	-0.140	27.965	0.000
		8	-0.136	-0.133	30.372	0.000
		9	-0.033	0.035	30.512	0.000
		10	-0.037	-0.001	30.692	0.000
		11	-0.060	0.023	31.175	0.001
		12	-0.308	-0.269	44.040	0.000
		13	-0.128	0.057	46.273	0.000
		14	0.036	0.105	46.449	0.000
		15	-0.012	-0.038	46.469	0.000
		16	-0.058	0.051	46.937	0.000
		17	-0.006	0.050	46.941	0.000

There seems to be some remaining autocorrelation in the residuals.

Now let's estimate an AR(2). In the equation specification type "p c ar(1) ar(2)" and check the residuals:

Equation: EQ2 Workfile: MF_3_4_9::W2_data\

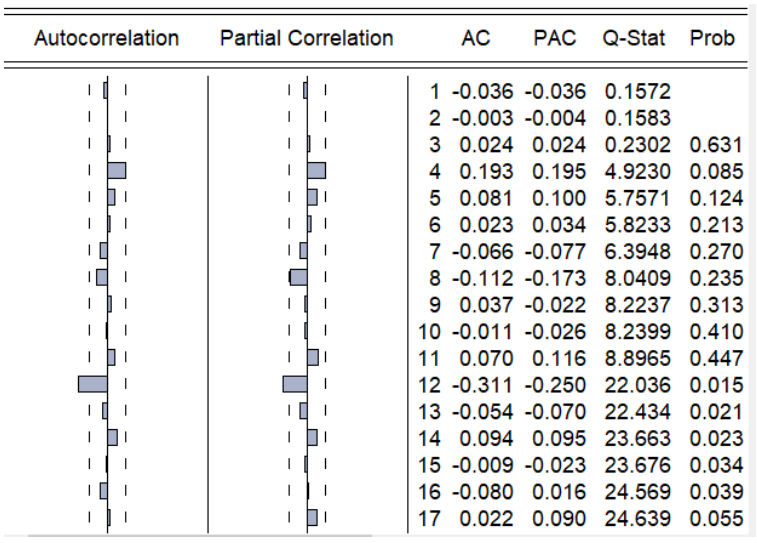
View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: P
Method: ARMA Maximum Likelihood (BFGS)
Date: 05/06/18 Time: 23:46
Sample: 2000M01 2009M12
Included observations: 120
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.571774	1.059932	6.200185	0.0000
AR(1)	1.321980	0.083122	15.90407	0.0000
AR(2)	-0.355405	0.080097	-4.437211	0.0000
SIGMASQ	0.241588	0.022000	10.98134	0.0000

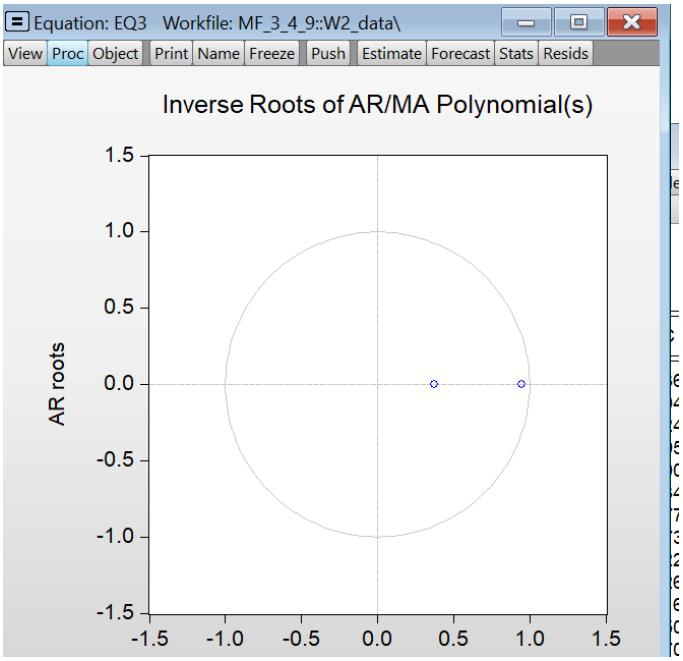
R-squared	0.955098	Mean dependent var	6.140182
Adjusted R-squared	0.953937	S.D. dependent var	2.329278
S.E. of regression	0.499918	Akaike info criterion	1.511453
Sum squared resid	28.99052	Schwarz criterion	1.604369
Log likelihood	-86.68718	Hannan-Quinn criter.	1.549187
F-statistic	822.4668	Durbin-Watson stat	2.042742
Prob(F-statistic)	0.000000		

Inverted AR Roots	.95	.38



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.036	-0.036	0.1572	
		2	-0.003	-0.004	0.1583	
		3	0.024	0.024	0.2302	0.631
		4	0.193	0.195	4.9230	0.085
		5	0.081	0.100	5.7571	0.124
		6	0.023	0.034	5.8233	0.213
		7	-0.066	-0.077	6.3948	0.270
		8	-0.112	-0.173	8.0409	0.235
		9	0.037	-0.022	8.2237	0.313
		10	-0.011	-0.026	8.2399	0.410
		11	0.070	0.116	8.8965	0.447
		12	-0.311	-0.250	22.036	0.015
		13	-0.054	-0.070	22.434	0.021
		14	0.094	0.095	23.663	0.023
		15	-0.009	-0.023	23.676	0.034
		16	-0.080	0.016	24.569	0.039
		17	0.022	0.090	24.639	0.055

The residuals seem closer to white noise now. Let's check that our estimated AR(2) is stationary:



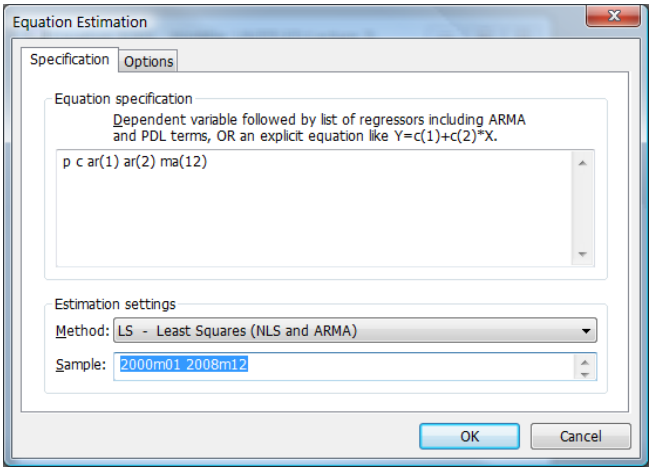
Indeed, the roots are within the unit circle.

Based on the correlograms and the residual diagnostics the AR(2) seems to be a better model.

Let's see what do the information criteria say? These are reported towards the end of the estimation output of each model.

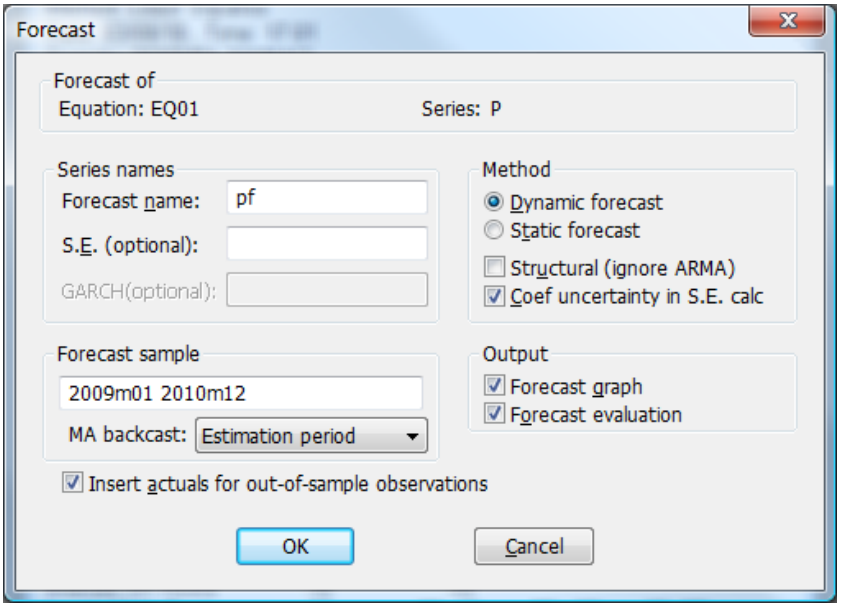
Model	AIC	HQ	BIC
AR(1)	1.628594	1.656895	1.698282
AR(2)	***1.511453***	***1.549187***	***1.604369***

Question 6. Use the preferred model structure from question 5 with an extra MA(12) component, and estimate it again for the subsample (why?) 2000M1–2008M12. Use the estimated model to produce monthly forecasts for 2009-2010. Compare these forecasts with actual inflation for 2009, and determine the root mean square forecasting error.

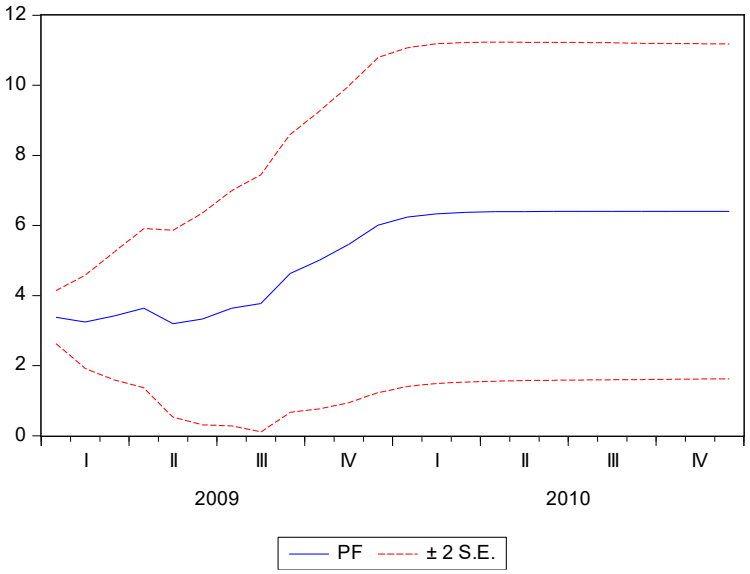


Specify the sample period in the equation specification window:

From the estimation window we can choose **Forecast**. In the window that will appear we can indicate that the forecast sample is 2009M1–2010M12. We will use the dynamic forecast method, which is the same method that was explained in the lecture.



As the forecast horizon increases the forecast approaches the mean of the series, and the standard errors become constant. Also, we can see that for the period 2009M1–2009M12 for which we have data, the forecast error is, on average, about 0.5 percentage point. The root mean squared error of the forecast is 0.65.



Forecast: PF	
Actual: P	
Forecast sample: 2009M01 2010M12	
Included observations: 24	
Root Mean Squared Error	0.646546
Mean Absolute Error	0.516121
Mean Abs. Percent Error	12.13185
Theil Inequality Coefficient	0.076363
Bias Proportion	0.046883
Variance Proportion	0.000444
Covariance Proportion	0.952673
Theil U2 Coefficient	1.226635
Symmetric MAPE	12.74117

Plotting the series P and PF for the period 2000M1–2010M12 (In the work file window select p, pf by pressing Ctrl key and click with mouse, right click, choose open as group, click View, choose graph and click ok) we can have a better sense of the forecasting performance of our model.

